

A man with a black headband is lying in a ball pit, with only his head and shoulders visible above the sea of colorful plastic balls. The balls are in various colors including red, yellow, blue, green, purple, and pink. In the background, several grey pillows are visible against a light-colored wall.

Starter activity

In as much detail as you can, describe how you might feel if you are buried at the bottom of a ball pit.

Pressure in Liquids and Gases

1. What is the equation for calculating speed?

speed = distance/ time

1. What is the equation for calculating moment?

moment = force x distance

1. Explain why it hurts if you stand on 1 nail but not lots of nails? (use the word pressure and surface area in your answer)

Because the surface area of many nails is much greater so the pressure on each nail is smaller

1. What 2 states of matter can flow?

Liquids and gases

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

Pressure in liquids and gases

- Liquids and gases are examples of **fluids**.
- Pressure in fluids acts in **all directions**.
- The particles in fluids are **moving** all the time and **colliding** with the walls of containers and other things they come into contact with.
- The **force of the collisions causes pressure**, which acts in all directions.
- **Pressure = Force/ Area**
- Pressure is measured in: **N/cm² or Pa**

Phet simulation

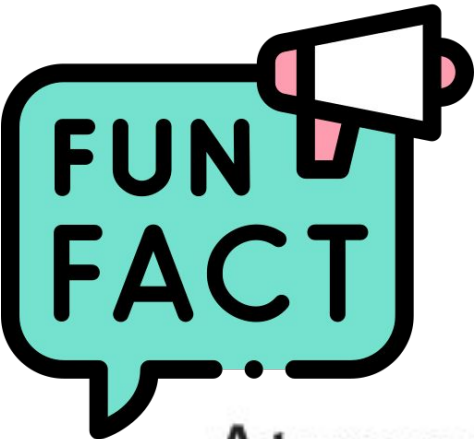
https://phet.colorado.edu/sims/html/gases-intro/latest/gases-intro_all.html

Go through the simulation above and complete these sentences in your book:

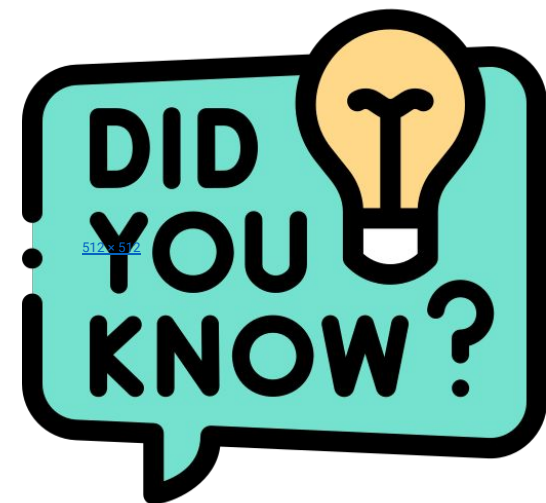
- Pressure in **fluids** (liquids and gases) are caused by the particles _____ with each other and the sides of the container.
- If the temperature increases the pressure ____.
- If the volume increases the pressure _____.
- If the number of particles increases the pressure _____.

Check your answer

- Pressure in **fluids** (liquids and gases) are caused by the particles **colliding** with each other and the sides of the container.
- If the temperature increases the pressure **increases**.
- If the volume increases the pressure **decreases**.
- If the number of particles increases the pressure **increases**.



Atmospheric pressure (the pressure of the air) at the surface of the Earth is about 100 000 pascals. One pascal (Pa) is a force of one newton on every square metre. We do not notice this pressure because fluids inside our bodies are at a similar pressure.

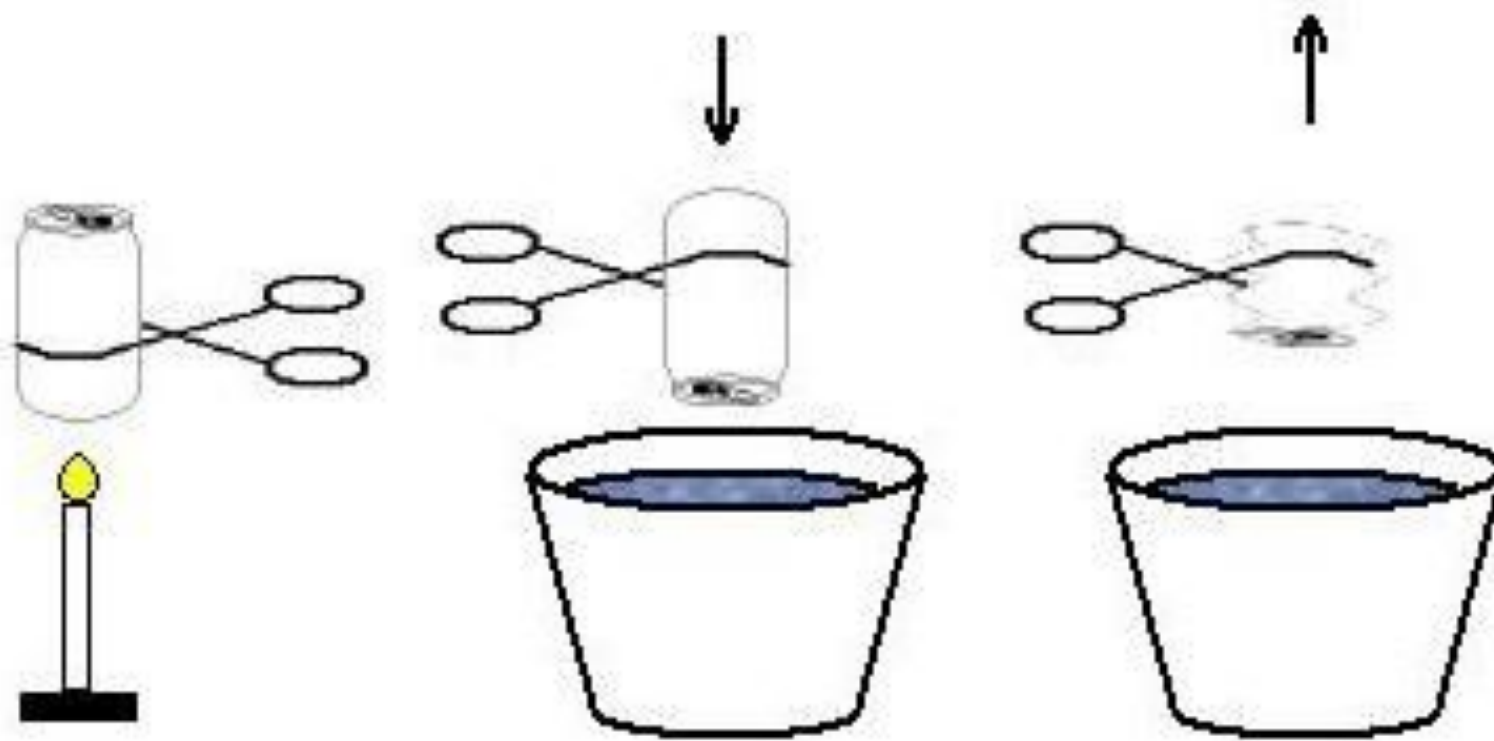


Something to think about:

1. What factors increase the pressure of fluids?

1. If you heat up a closed can of coke, what might happen?

Demo! Crushing Can Experiment



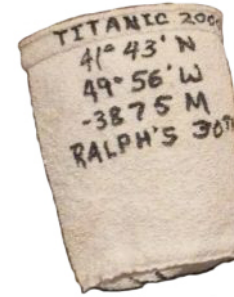
The "crushing can experiment" demonstrates the power of **atmospheric pressure** by heating water in a soda can until **steam pushes the air out**, then quickly inverting it into cold water. The **rapid cooling causes the steam to condense** into a small amount of liquid, creating a **vacuum** inside. The **external air pressure, now much greater than the pressure inside the can, crushes it instantly**.

Effects of Pressure in Fluids

An expedition recently took Styrofoam cups down to the Titanic.



At sea
level



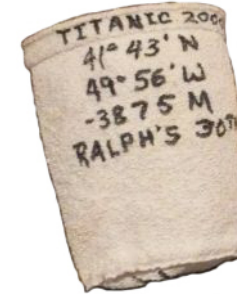
At 3800 metres below sea
level

How do you explain the difference between the Styrofoam cups?

At sea level



At 3800 metres below sea level



The cups taken down to the wreck have been crushed.
This is due to the water pressure at 3800 metres being much greater than the atmospheric pressure at sea level..

Can you explain what you see here?



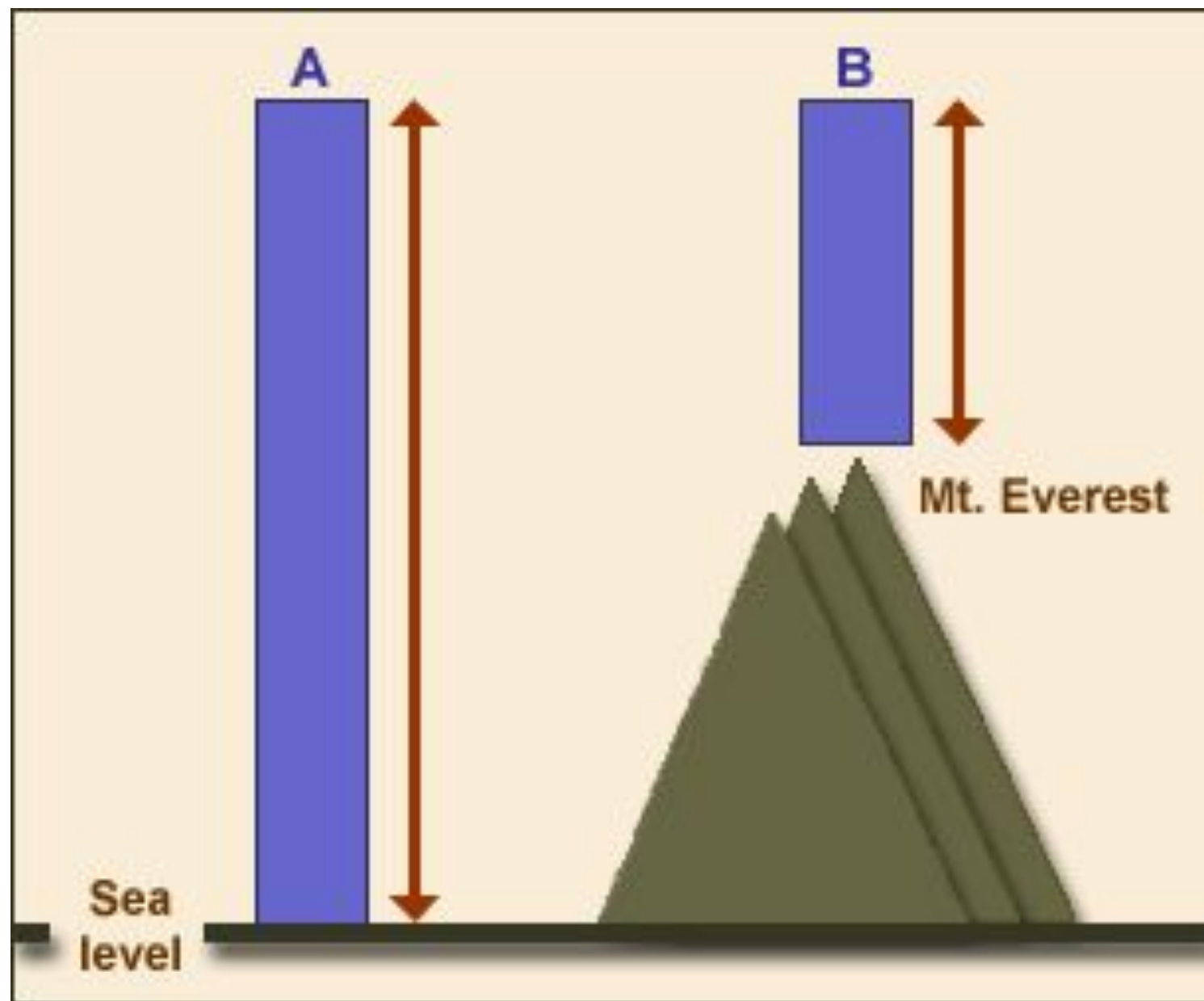
This photo was taken at a height of 2413 m.



This is the same bag of crisps at sea level.

E | These photos demonstrate the effect of air pressure on a sealed bag of crisps.

Can you suggest anything similar you might see when you've been on a plane?



Pressure depends on the weight of fluid above.

At the bottom of the ocean there is more water above you so greater pressure.

At the top of a mountain there is less air above you so lower pressure.



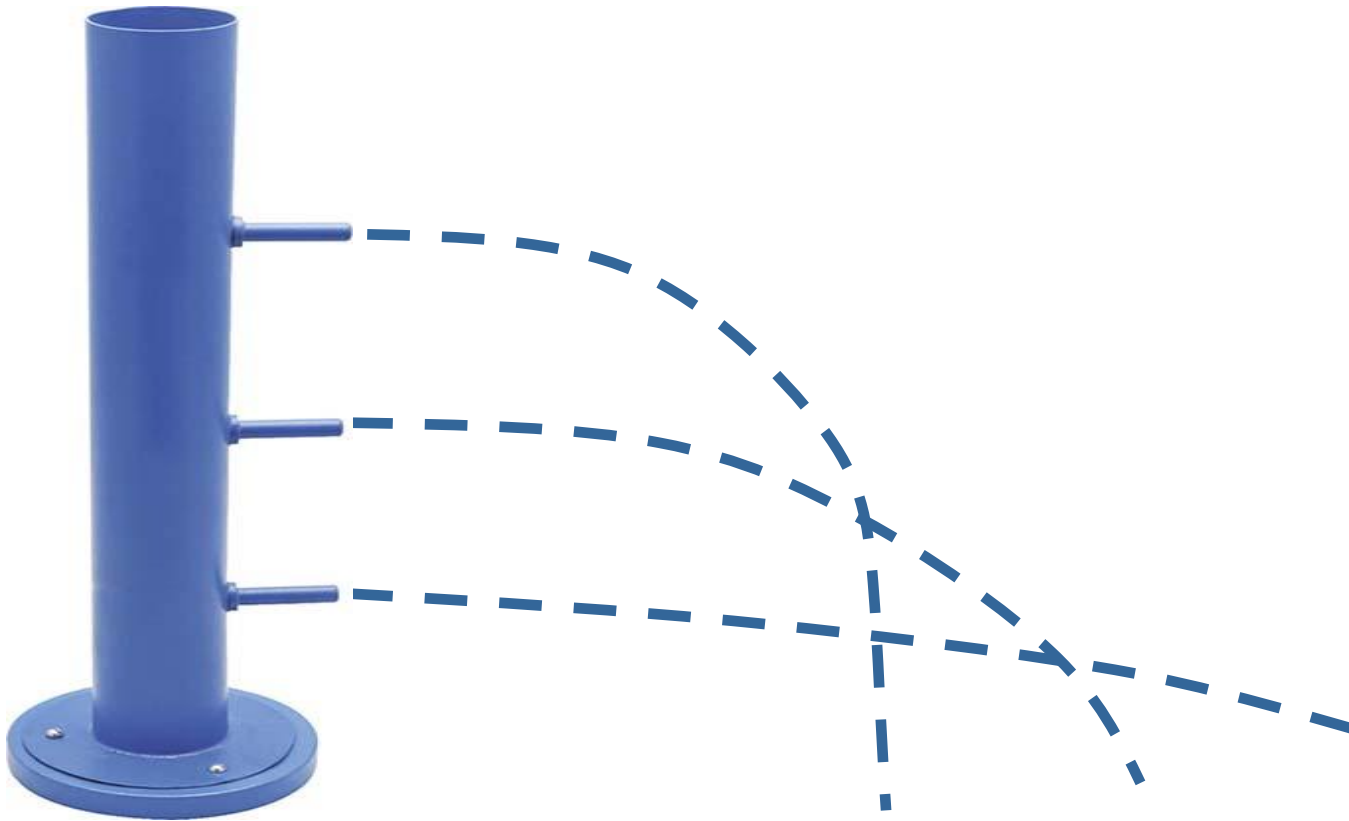
A | People diving for fun usually stay within 30 m of the surface. People who work further below the sea need special suits, like this one, to stop the high pressure from the water harming them.



D | Mountaineers need to carry tanks of oxygen with them to help them to breathe at high altitude. Without oxygen, they may suffer from altitude sickness and could die.

Pressure vs. Depth

What does this demonstration tell you?

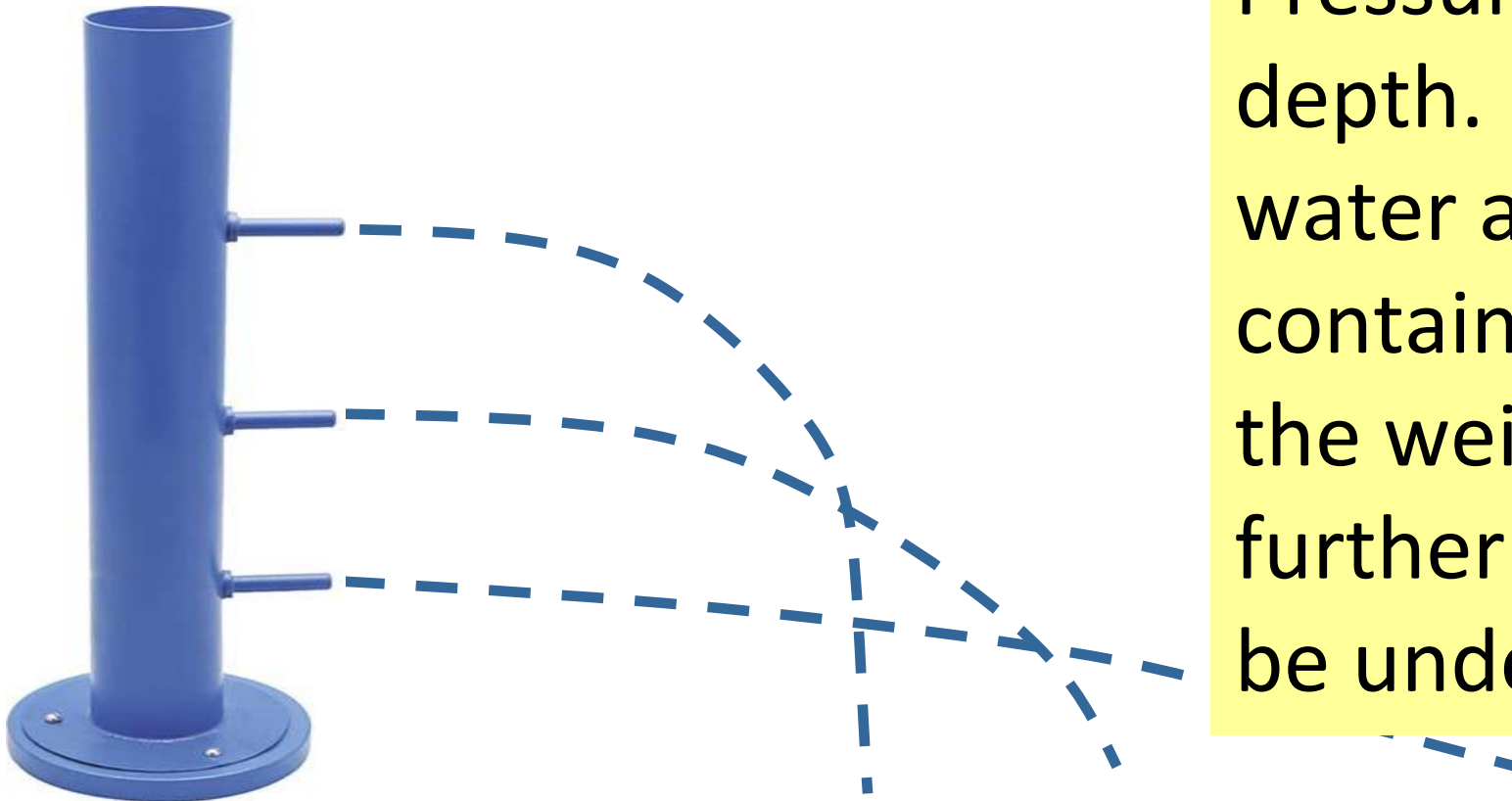


Pressure increases with _____. This is because the water at the _____ of this container is pushed on by the _____ of the water further up, which causes it to be under higher _____.

Words – pressure, bottom, weight, depth

Pressure vs. Depth

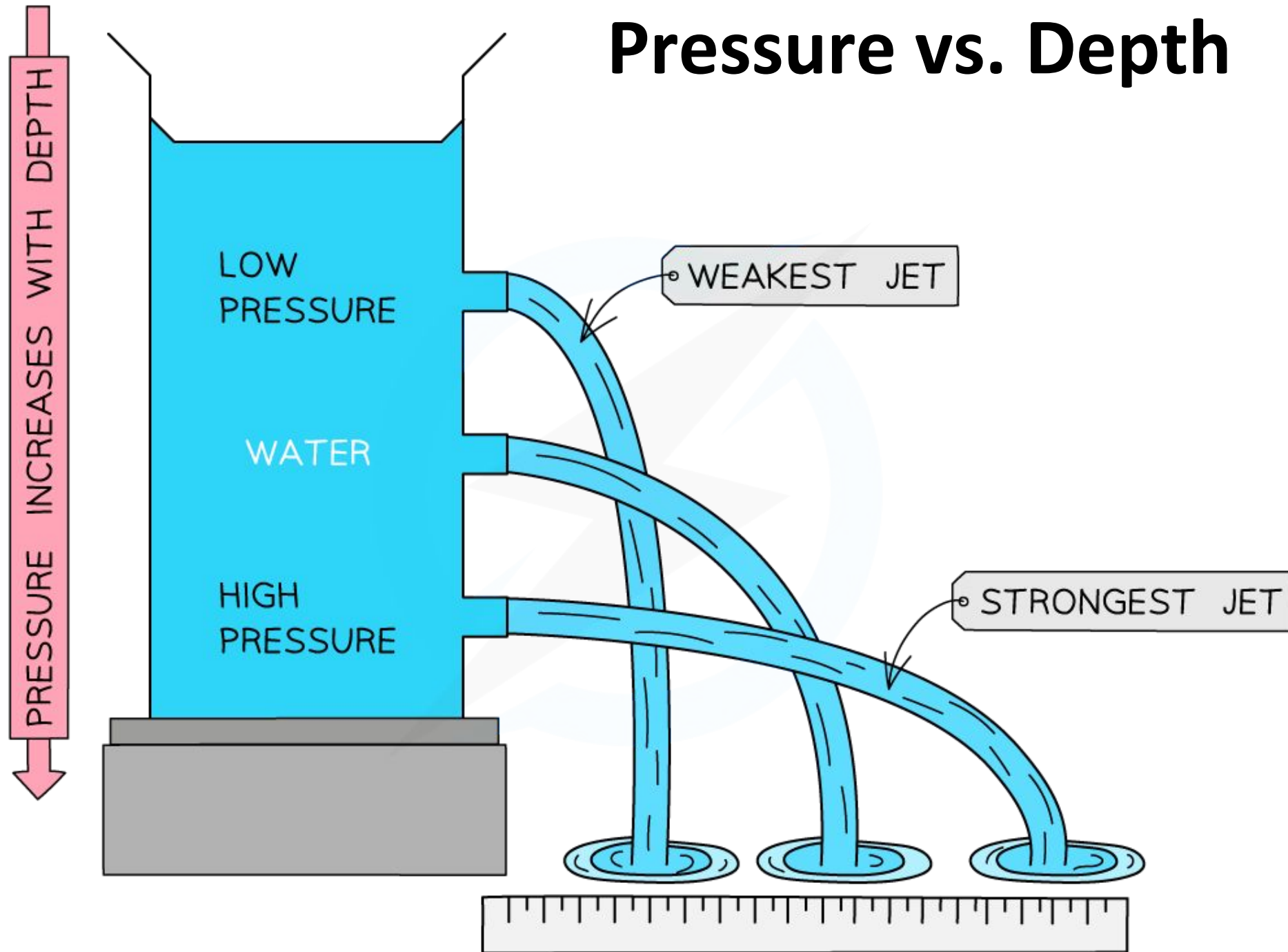
What does this demonstration tell you?



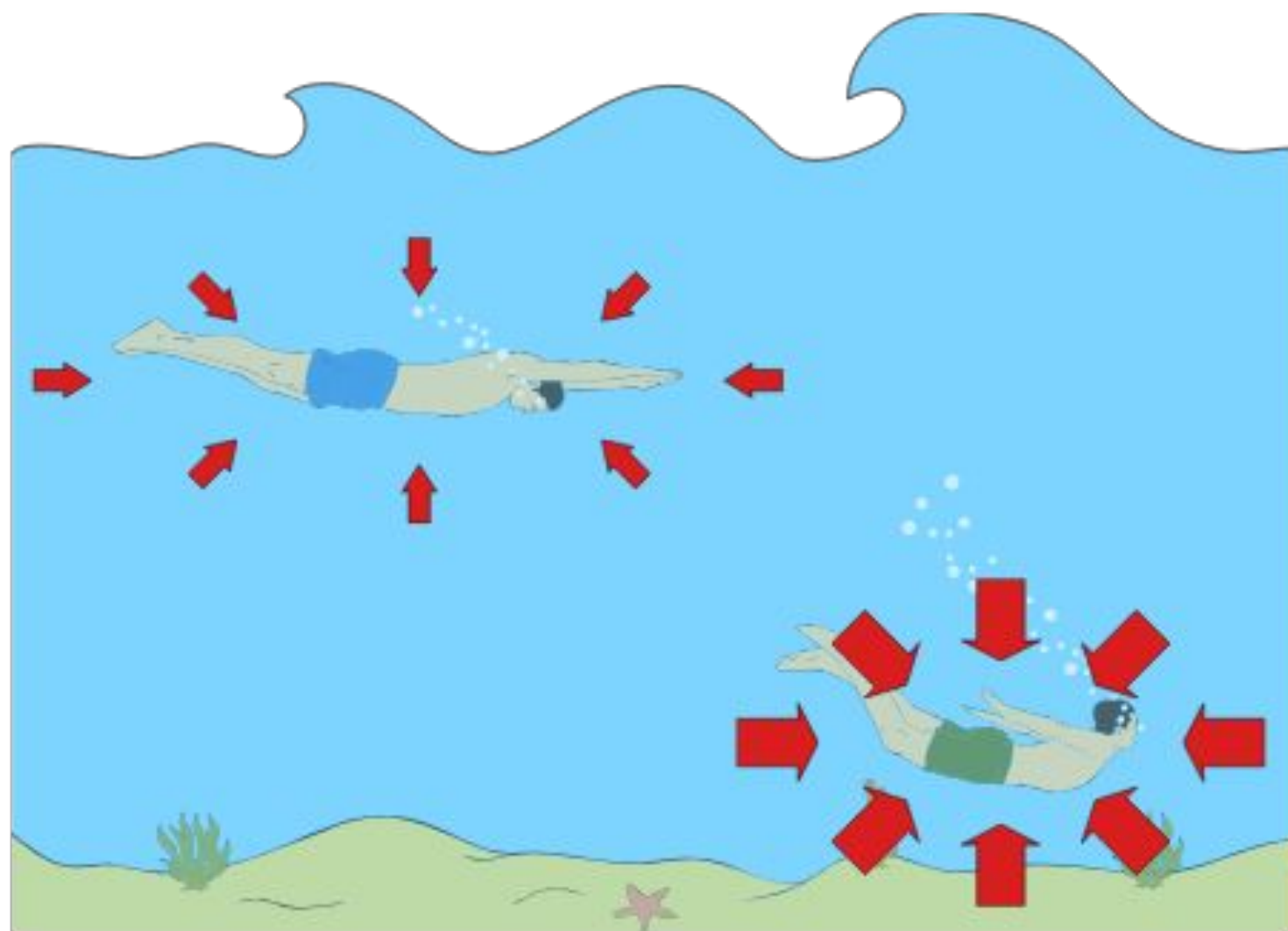
Pressure increases with depth. This is because the water at the bottom of this container is pushed on by the weight of the water further up, which causes it to be under higher pressure.

Words – pressure, bottom, weight, depth

Pressure vs. Depth



Water surface



Increasing depth

1 Fluids are:

- ☐ only liquids
- ☐ only gases
- ☐ only solids
- ☐ liquids or gases.

2 Pressure in a fluid is caused by:

- the forces from particles hitting the container
- a heavy object
- having a small area.

3 Car tyres have high pressure inside them because:

- ☐ they have more air particles inside them
- ☐ they have fewer air particles inside them
- ☐ they are made from rubber

4 Air pressure:

- gets less as you go higher
- gets less as you go deeper
- gets bigger as you go higher.

5 Pressure in water:

- gets less as you go higher
- gets less as you go deeper
- gets bigger as you go deeper

This table shows how air pressure decreases with height above the Earth.

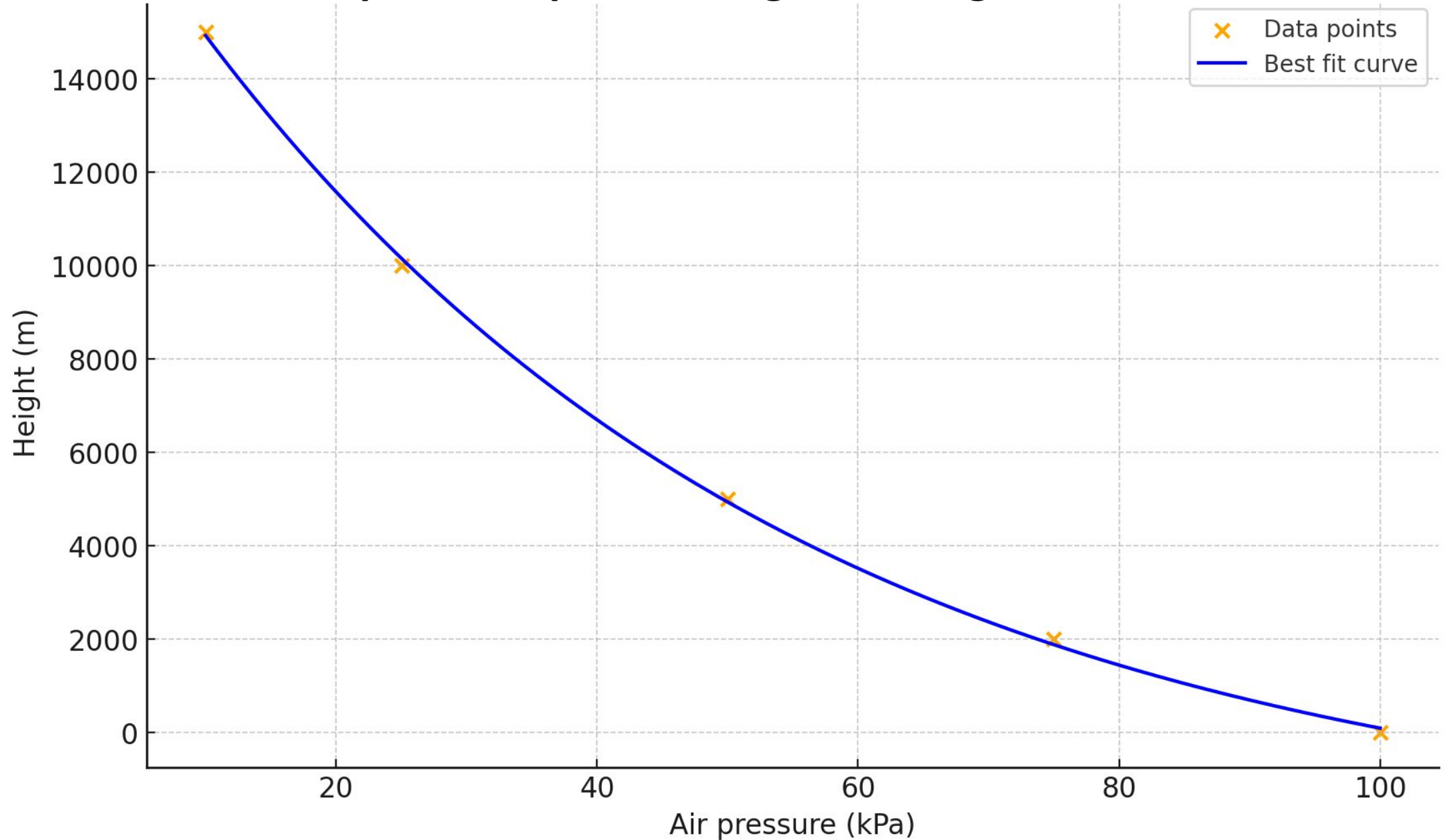
a) Plot a graph on graph paper to show this information. Put air pressure on the horizontal axis and height on the vertical axis. Your scale for the height axis can go up in 2000s, from 0 to 16 000 metres. Don't forget to give your graph a title.

a) Use your graph to find the air pressure at 3000 m.

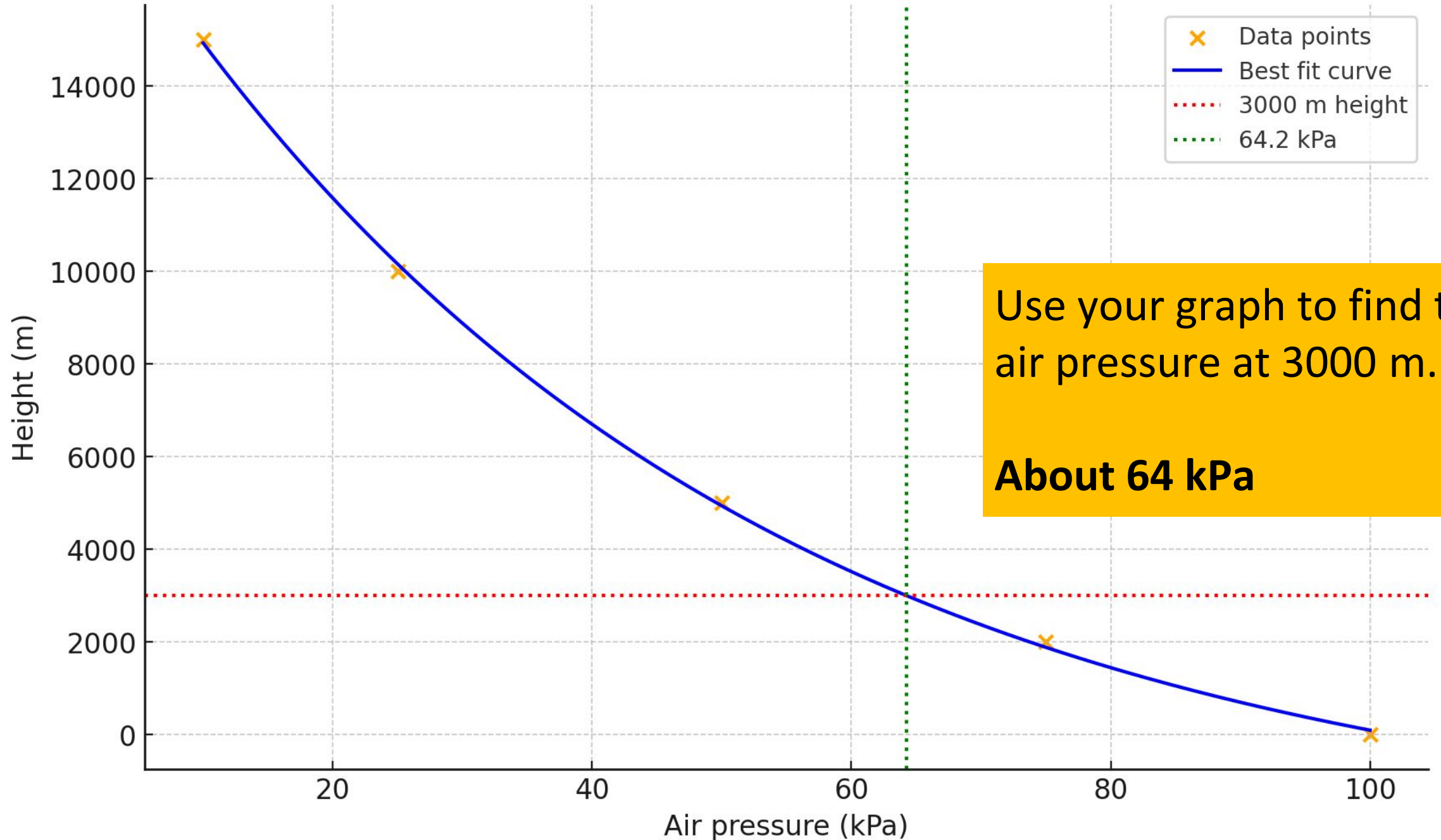
Height (m)	Air pressure (kPa)
0	100
2000	75
5000	50
10 000	25
15 000	10



Graph of Air pressure against Height

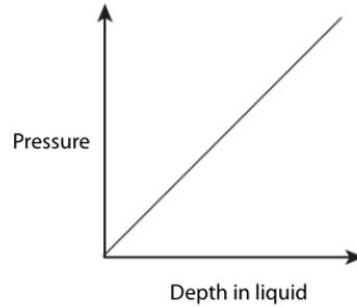


Air Pressure vs Height (Finding Air Pressure at 3000 m)



Questions

- 1
 - a Submarines are ships which are designed to go underwater. Explain why submarines that go to the deepest parts of the oceans must be very strong.
 - b Explain why the wall of a dam is thicker at the bottom than at the top.
- 2 The sketch graph shows how the pressure in a liquid changes with depth in the liquid.
 - a Explain the pattern shown in the graph.
 - b Copy and complete the sentence using the best word. Use information in the graph.
When the depth in the liquid doubles, the pressure in the liquid
- 3 A fish is in water. The water exerts pressure on the fish. Which of these causes pressure on the fish? Write **one** letter.
 - A The weight of water beside the fish.
 - B The weight of water all around the fish.
 - C The weight of water above the fish.
 - D The weight of water below the fish.
- 4 Marcus plays basketball. The ball is filled with air.
 - a Explain what causes the pressure inside the ball.
 - b Explain how putting more air in the ball will change the pressure inside the ball.
 - c Marcus notices that the pressure inside the ball is lower on a cold day and higher on a hot day.
Explain why the pressure inside the ball is higher on a hot day.
- 5 A metal container is connected to a vacuum pump. The vacuum pump removes all the air from inside the metal container. There is now a vacuum inside the metal container.
 - a Explain why the pressure in a vacuum is zero.
 - b The metal container collapses when there is a vacuum inside. Explain why.



Extension (if have extra time)
Learner's book page 110-111

Answers will be discussed after the timer ends.

10:00

Questions

- 1 a Pressure increases with depth in water; submarines must be very strong to withstand this pressure / not be damaged by the pressure.
- b Pressure increases with depth in water; the wall needs to be stronger at the bottom than at the top to withstand the greater pressure.
- 2 a Pressure increases with depth / pressure is directly proportional to depth.
- b When the depth in the liquid doubles, the pressure in the liquid **doubles**.
- 3 C

Mark your work using a green pen!

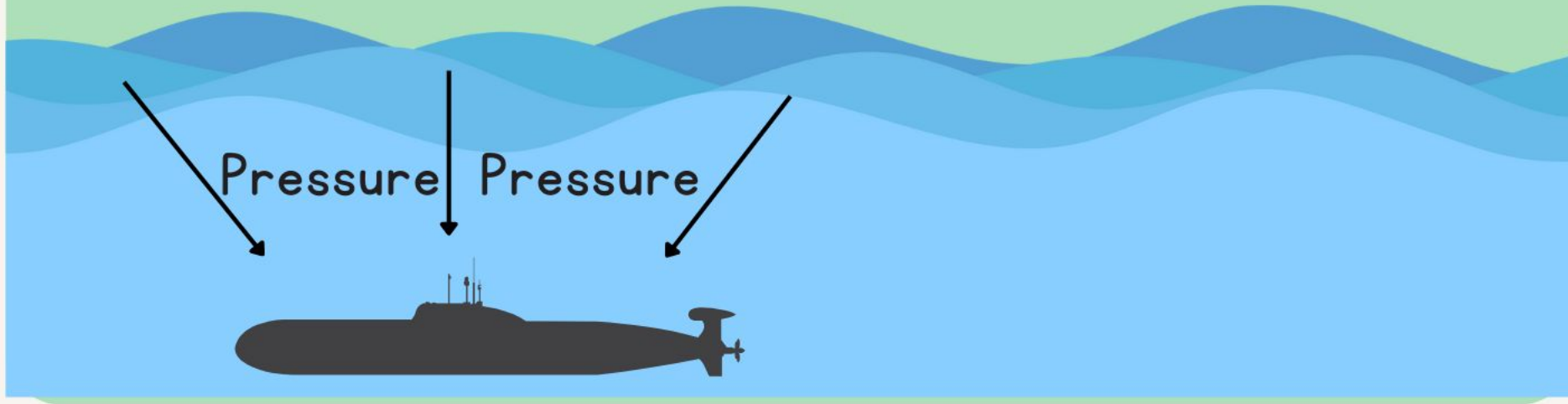
- 4** **a** Particles in the air inside the ball colliding with the inside surface of the ball.
- b** More particles inside the ball, so more collisions with the inside surface.
- c** Particles move faster at higher temperature; collisions with the inside surface have more force and occur more frequently.
- 5** **a** There are no particles to collide.
- b** Particles in the air collide with the outside of the container; there are no particles colliding with the inside of the container; the forces on the container are not balanced and the force from the outside is greater.

Mark your work using a green pen!

PRESSURE

We experience pressure when we are: **UNDER WATER**

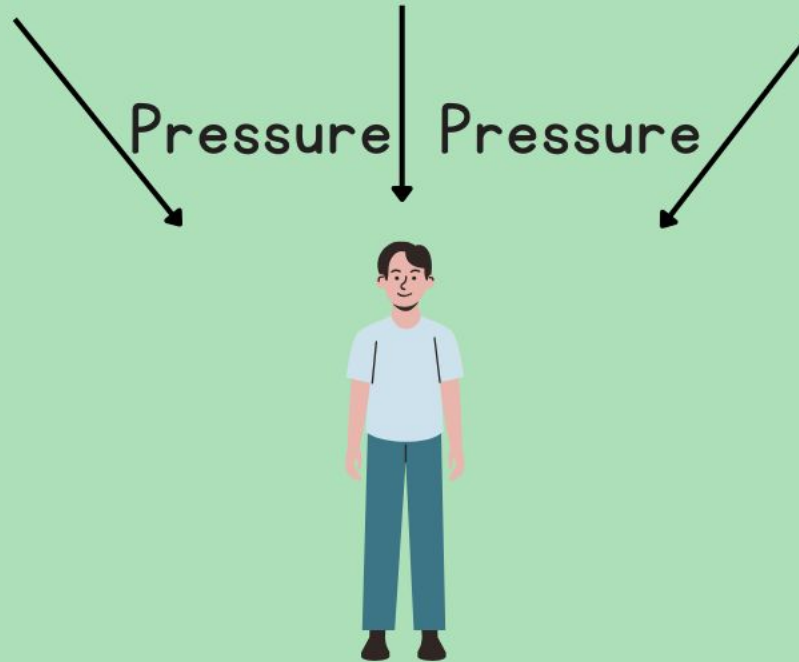
Submersion underwater results in pressure due to the weight of water above pressing down on any submerged object. Deeper depths lead to increased water pressure.



PRESSURE

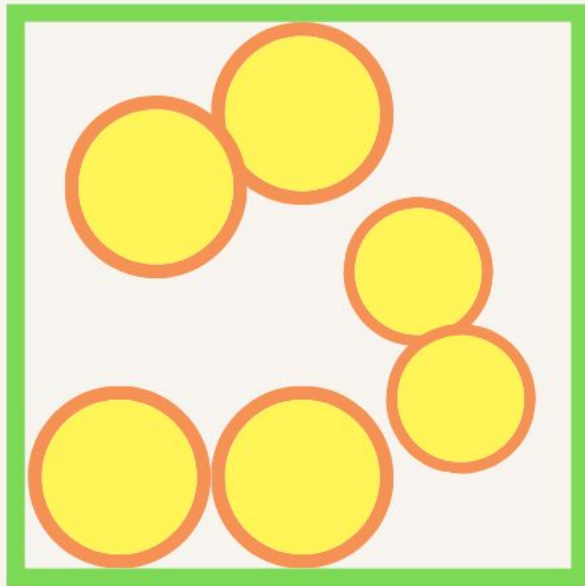
We experience pressure when we are: **ON EARTH**

We experience atmospheric pressure (about 101325 Pa)

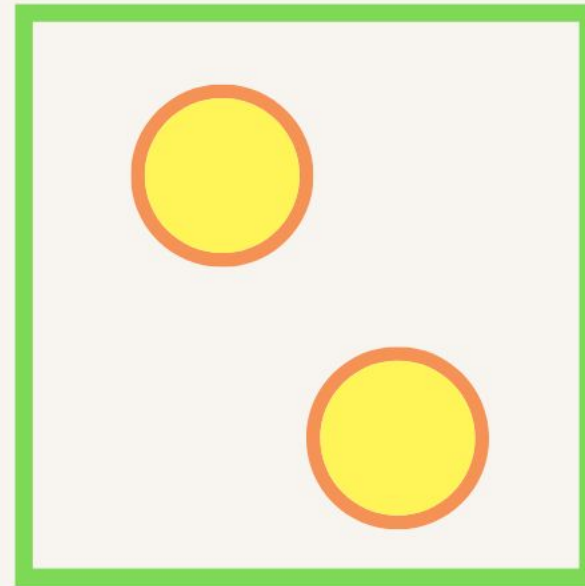


PRESSURE

Water exerts significantly higher pressure compared to air because water is much denser than air.



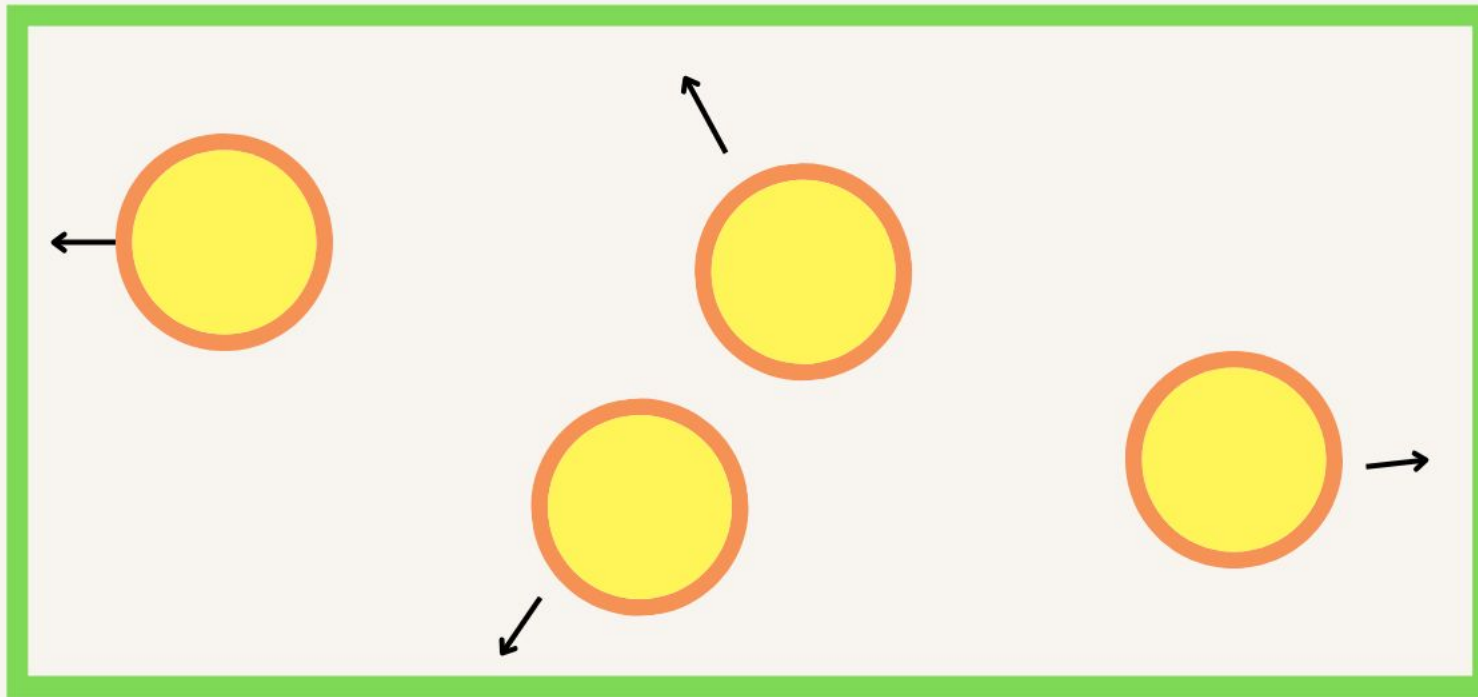
Water



Air

PRESSURE

Pressure arises from the impact of molecules colliding with every surface they encounter.



CALCULATING PRESSURE

Pressure is the force per unit area acting on a surface.

“

EQUATION

$$P = F / A$$

”

“

UNITS:

N / M²
PASCAL

”



INTUITION



“

EQUATION

$$P = F / A$$

”



WORKED EXAMPLE

Which exerts greater pressure: a force of 300 N acting on 0.5 m², or the same force acting on 1.0 m² ?





WORKED EXAMPLE

Which exerts greater pressure: a force of 300 N acting on 0.5 m², or the same force acting on 1.0 m² ?

$$P = F / A$$

$$= 300 / 0.5$$

$$= 600 \text{ Pa}$$

$$P = F / A$$

$$= 300 / 1$$

$$= 300 \text{ Pa}$$

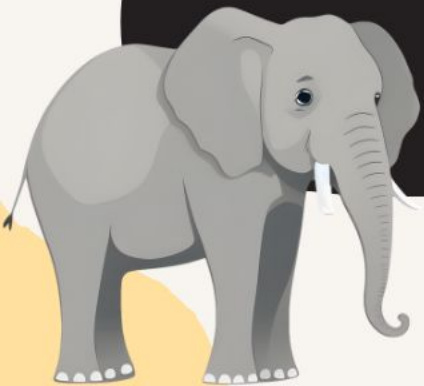
Answer:

A force acting on smaller area has greater pressure.



WORKED EXAMPLE

What pressure is exerted by an elephant weighing
60000 N acting on 1.2 m^2 ?





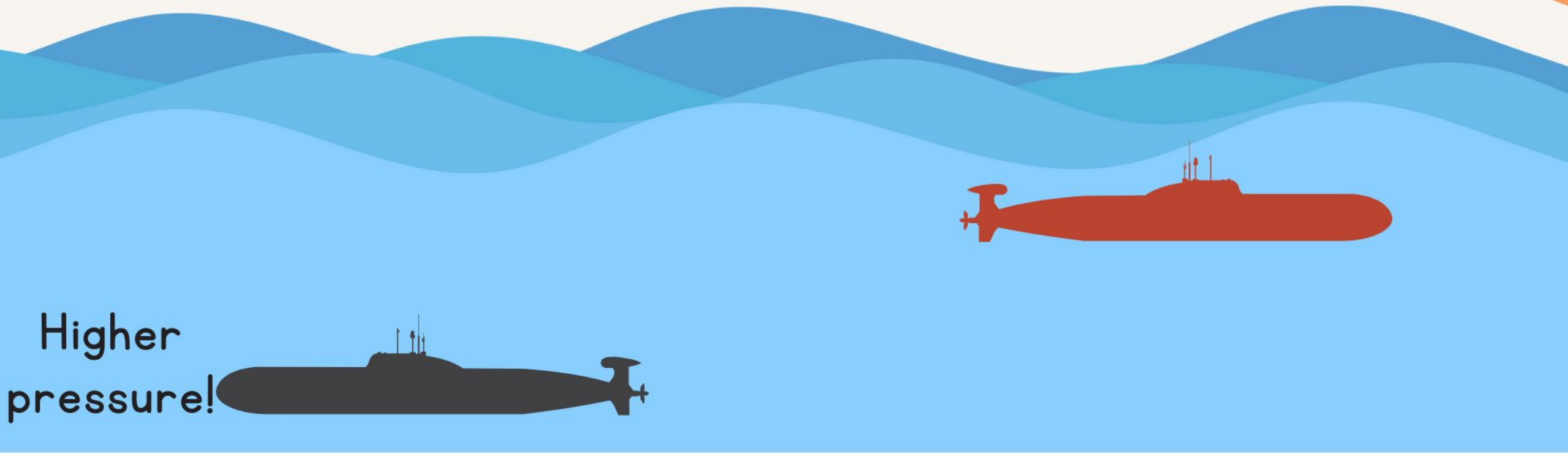
WORKED EXAMPLE

What pressure is exerted by an elephant weighing
60000 N acting on 1.2 m²?

$$\begin{aligned} P &= F / A \\ &= 60000 / 1.2 \\ &= 50000 \text{ Pa} \end{aligned}$$

CALCULATING PRESSURE

We have seen that the deeper one dives into water, the greater the pressure.

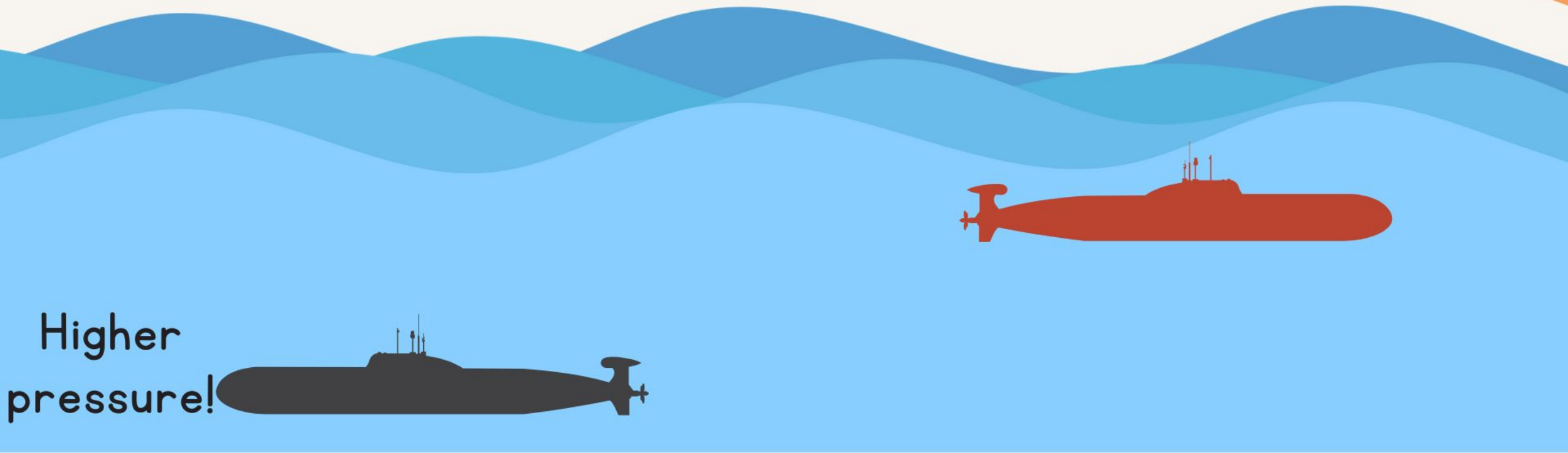


Higher
pressure!

CALCULATING PRESSURE

Pressure under liquid depends on:

- a. Depth (we use the letter h , for height)
- b. Density (we use the letter ρ)



CALCULATING PRESSURE

Equations for pressure under liquid:

“

EQUATION

”

$$\text{PRESSURE} = \rho g h$$

Height

”

Density of liquid

Gravitational
field strength



WORKED EXAMPLE

Determine the pressure at the base of a lake that is 3.0 meters in depth.

How does this pressure compare with atmospheric pressure, which is 101, 325 Pa? (Density of water is 1000 kg/m^3).





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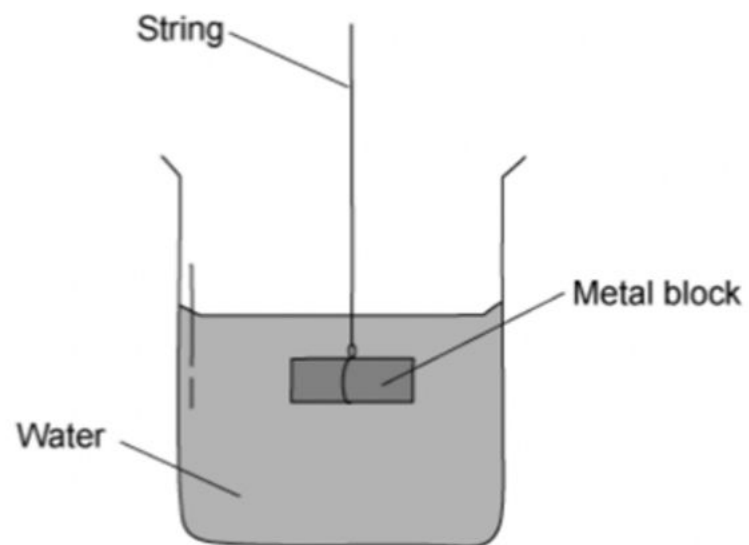
$$\begin{aligned}\text{Pressure} &= pgh \\ &= 1000 (9.8) (3) \\ &= 29400 \text{ Pa}\end{aligned}$$

The atmospheric pressure is larger than pressure under the lake.

PYQ

A small metal block is suspended under the surface of a beaker of water by a string.

The metal block experiences a pressure exerted by the liquid.



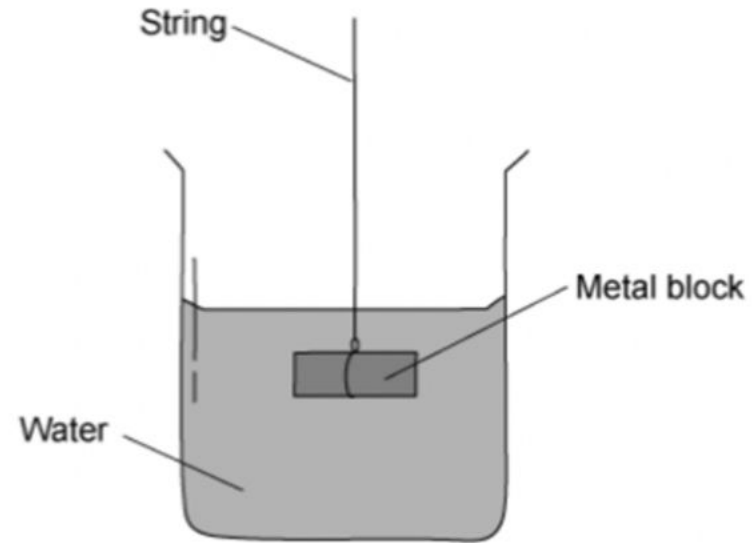
What would increase the pressure exerted on the metal block?

- A** increasing the surface area of the stone
- B** using a liquid with a lower density
- C** increasing the mass of the metal block
- D** lowering the metal block deeper into the liquid

PYQ

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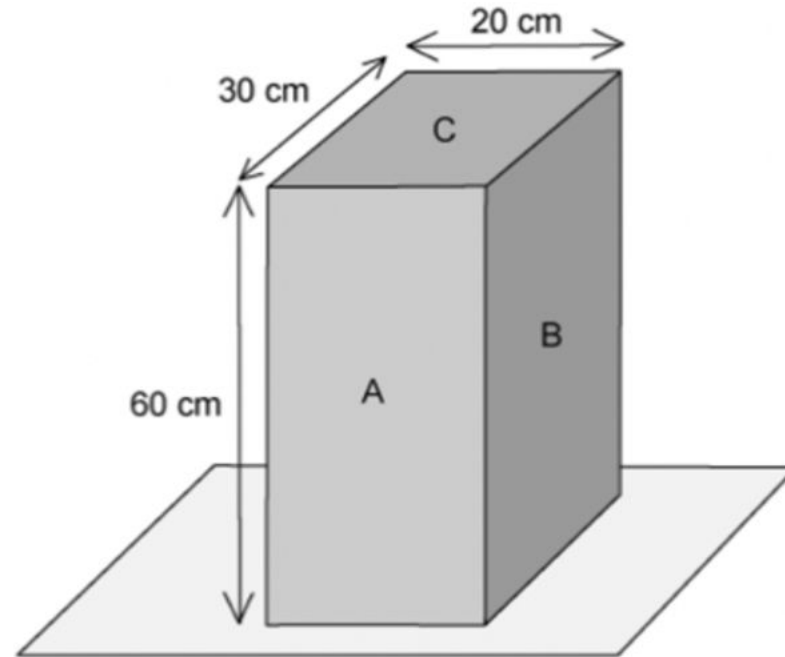


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PYQ

A wooden block rests on a table.

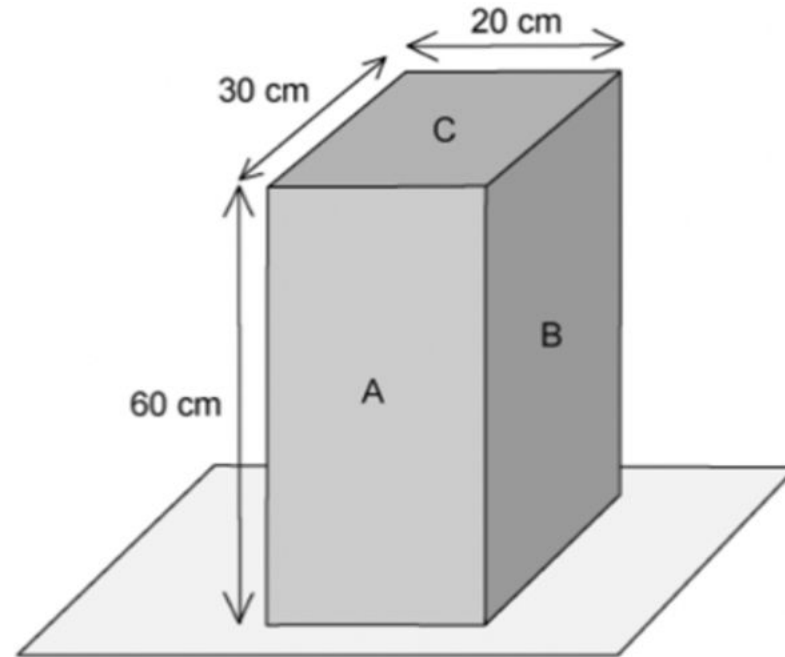


On which surface should the block be laid to produce the largest pressure on the table?

- A** A.
- B** B
- C** C
- D** Any side, they will all produce the same pressure

PYQ

A wooden block rests on a table.



On which surface should the block be laid to produce the largest pressure on the table?

A A.

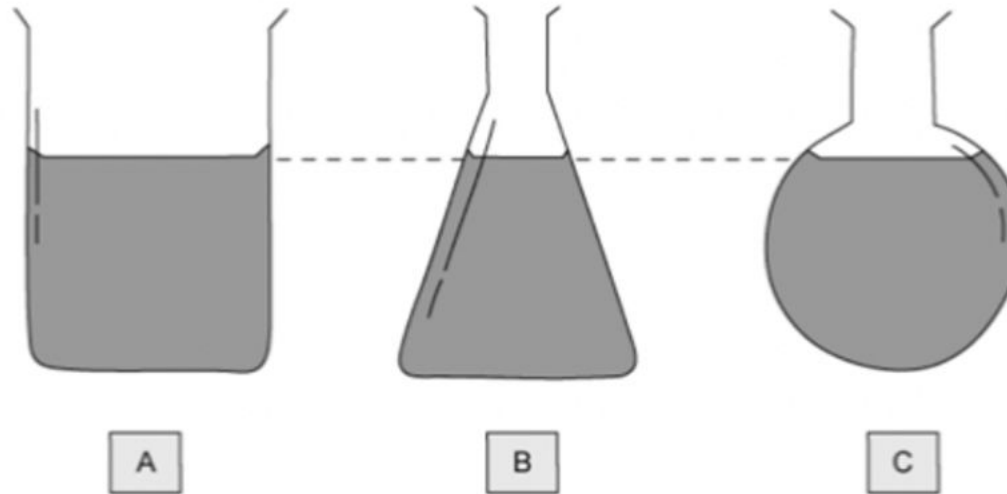
B B

C C

D Any side, they will all produce the same pressure

PYQ

Three beakers of water are placed on a table. The depth of water in each container is the same.

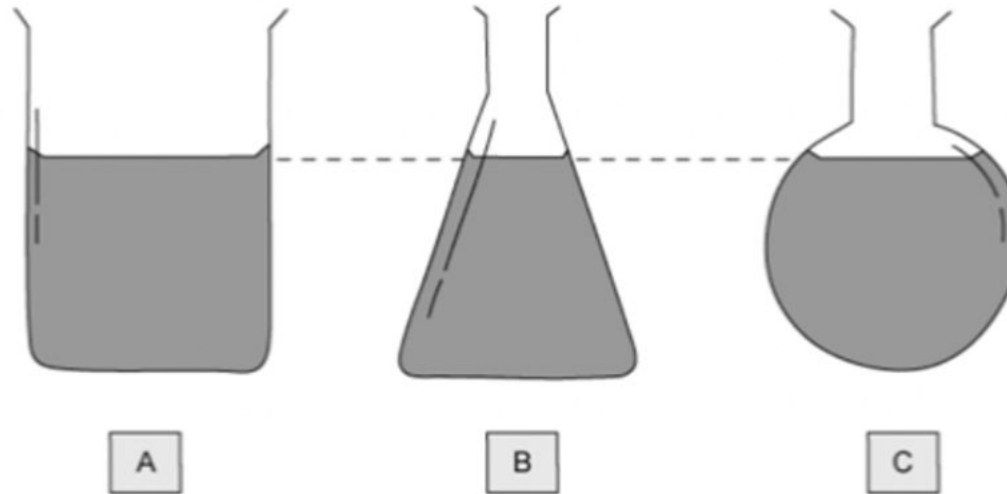


In which container does the water exert the greatest pressure on the base of the container?

- A A
- B B
- C C
- D None, the pressure is the same in all three.

PYQ

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In which container does the water exert the greatest pressure on the base of the container?

A A

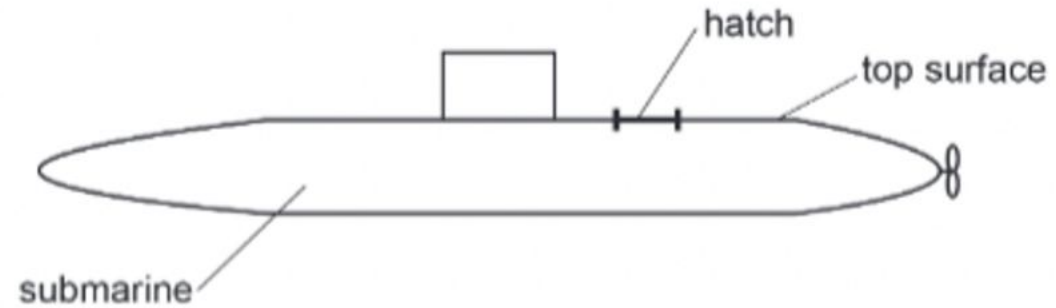
B B

C C

D None, the pressure is the same in all three.

PYQ

- 1 The diagram shows a submarine. The submarine is fully submerged in the sea.



The density of sea water is 1020 kg/m^3 .

The atmospheric pressure is 100 kPa and the total pressure on the top surface of the submarine is 500 kPa .

Calculate the depth of the top surface of the submarine below the surface of the sea.

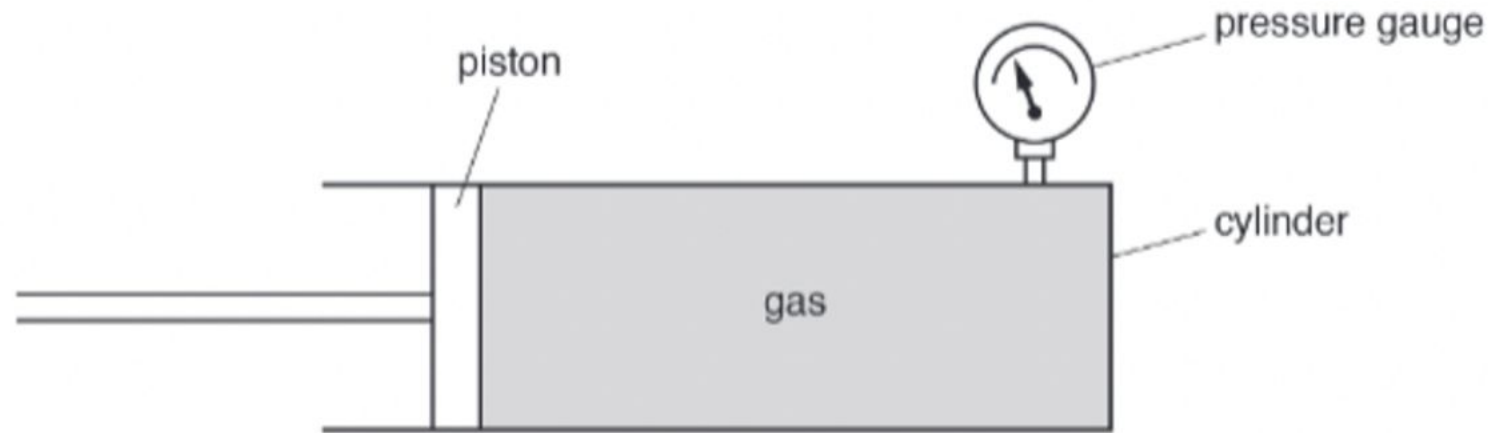
depth = [3]

PYQ

Answer	Marks
$P = \rho gh$ in any form (1) (h =) $400 \times 10^3 / (1020 \times 10)$ (1) (h =) 39 m (1)	3

PYQ

The diagram shows a gas contained in a cylinder enclosed by a piston.



Describe, in terms of momentum of the molecules, how a pressure is exerted on the walls of the cylinder.

.....

.....

.....

.....

.....

PYQ

Answer	Marks
they / molecules collide with <u>walls</u> (1) <u>change of momentum</u> causes <u>force</u> (to be exerted on walls) (1) pressure = force / area (so pressure is exerted on walls) (1)	3